

# Experiment SINGLE\_BIT on password\_check.o with target Target.X86\_64

## Settings

- **program\_file:** /home/dblshot/ghPackages/FaultSim/experiments/para\_bit\_x86\_64\_password\_check/password\_check.o
- **out\_dir:** /home/dblshot/ghPackages/FaultSim/experiments/para\_bit\_x86\_64\_password\_check/mutated\_bins
- **program\_input:**['w', 'r', 'o', 'n', 'g', '\n']
- **expected\_stdout:**['C', 'o', 'r', 'r', 'e', 'c', 't']
- **expected\_returncode:** 0
- **list\_expected:** False
- **timeout:** 4
- **save\_results:** /home/dblshot/ghPackages/FaultSim/experiments/para\_bit\_x86\_64\_password\_check/results.csv
- **yes:** True

## Binary information + Running the binary information

- Contains **123** instructions
- Therefore, FaultSim attempted to make **123** mutations
- Of the **123** attempted mutations, **3007** valid mutated binaries were generated
- The target arch was Target.X86\_64
- The compile command was: gcc test\_files/password\_check.c -o experiments/para\_bit\_x86\_64\_password\_check/password\_c
- An example run command: timeout 4s /home/dblshot/ghPackages/FaultSim/experiments/para\_bit\_x86\_64\_password\_check/
- The NOP for this binary is: Nop.X86\_64
- The runtime to generate and run binaries was: 0:00:10.129862

## Return Code Frequencies

| <i>Exitcode</i>       | <i>Frequency</i> |
|-----------------------|------------------|
| password denied (97)  | 1841             |
| EX SIGSEGV (139)      | 936              |
| EX SIGABRT (134)      | 121              |
| password accepted (0) | 43               |
| Exit 0 : Bad STDOUT   | 1                |
| EX SIGILL (132)       | 32               |
| failed to run (-900)  | 8                |
| 133                   | 7                |
| 135                   | 4                |
| 6                     | 3                |
| 124                   | 2                |
| 7                     | 1                |
| 225                   | 1                |
| 33                    | 1                |
| EX DATAERR (65)       | 1                |
| 113                   | 1                |
| 105                   | 1                |
| 101                   | 1                |
| 99                    | 1                |
| 96                    | 1                |

## Programs that ran critical code

**43** programs ran the critical code out of **3007** mutated binaries. The binaïres were: - password\_check.o\_0x123d\_24 - password\_check.o\_0x123d\_25 - password\_check.o\_0x123d\_26 - password\_check.o\_0x123d\_27 - password\_check.o\_0x123d\_28 - password\_check.o\_0x123d\_29 - password\_check.o\_0x123d\_30 - password\_check.o\_0x123d\_31 - password\_check.o\_0x1248\_29 - password\_check.o\_0x1248\_31 - password\_check.o\_0x12a0\_19 - password\_check.o\_0x12a3\_19 - password\_check.o\_0x12a3\_23 - password\_check.o\_0x12ab\_10 - password\_check.o\_0x12ab\_11 - password\_check.o\_0x12ab\_13 - password\_check.o\_0x12ab\_14 - password\_check.o\_0x12ad\_7 - password\_check.o\_0x12ad\_13 - password\_check.o\_0x12a6\_9 - password\_check.o\_0x12c9\_1 - password\_check.o\_0x12c9\_14 - password\_check.o\_0x12c9\_16 - password\_check.o\_0x12c9\_17 - password\_check.o\_0x12c9\_18 - password\_check.o\_0x12c9\_19 - password\_check.o\_0x12c9\_20 - password\_check.o\_0x12c9\_21 - password\_check.o\_0x12c9\_22 - password\_check.o\_0x12c9\_23 - password\_check.o\_0x12c9\_24 - password\_check.o\_0x12c9\_25 - password\_check.o\_0x12c9\_26

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- password_check.o_0x12c9_27 - password_check.o_0x12c9_28 - password_check.o_0x12c9_29 - password_check.o_0x12c9_30
- password_check.o_0x12c9_31 - password_check.o_0x12cd_1 - password_check.o_0x12cd_4 - password_check.o_0x12cd_5 -
password_check.o_0x12cd_7 - password_check.o_0x12e5_28
```

### Program 0 password\_check.o\_0x123d\_24 diassembleby vs vanilla

|                               |                                   |                             |                                   |
|-------------------------------|-----------------------------------|-----------------------------|-----------------------------------|
| 0 0x1233 b8 00 00 00 00       | mov eax, 0                        | 0x1233 b8 00 00 00 00       | mov eax, 0                        |
| 1 0x1238 e8 a3 fe ff ff       | call 0x10e0                       | 0x1238 e8 a3 fe ff ff       | call 0x10e0                       |
| 2 0x123d c6 45 ec 00          | mov byte ptr [rbp - 0x14], 0      | 0x123d c6 45 ec 80          | mov byte ptr [rbp - 0x14], 0      |
| 3 0x1241 48 8b 15 c8 2d 00 00 | mov rdx, qword ptr [rip + 0x2dc8] | 0x1241 48 8b 15 c8 2d 00 00 | mov rdx, qword ptr [rip + 0x2dc8] |

### Program 1 password\_check.o\_0x123d\_25 diassembleby vs vanilla

|                               |                                   |                             |                                   |
|-------------------------------|-----------------------------------|-----------------------------|-----------------------------------|
| 0 0x1233 b8 00 00 00 00       | mov eax, 0                        | 0x1233 b8 00 00 00 00       | mov eax, 0                        |
| 1 0x1238 e8 a3 fe ff ff       | call 0x10e0                       | 0x1238 e8 a3 fe ff ff       | call 0x10e0                       |
| 2 0x123d c6 45 ec 00          | mov byte ptr [rbp - 0x14], 0      | 0x123d c6 45 ec 40          | mov byte ptr [rbp - 0x14], 0      |
| 3 0x1241 48 8b 15 c8 2d 00 00 | mov rdx, qword ptr [rip + 0x2dc8] | 0x1241 48 8b 15 c8 2d 00 00 | mov rdx, qword ptr [rip + 0x2dc8] |

## Program 2 password\_check.o\_0x123d\_26 diassemblebly vs vanilla

|          |                      |                                   |        |                      |                     |
|----------|----------------------|-----------------------------------|--------|----------------------|---------------------|
| 0 0x1233 | b8 00 00 00 00       | mov eax, 0                        | 0x1233 | b8 00 00 00 00       | mov eax, 0          |
| 1 0x1238 | e8 a3 fe ff ff       | call 0x10e0                       | 0x1238 | e8 a3 fe ff ff       | call 0x10e0         |
| 2 0x123d | c6 45 ec 00          | mov byte ptr [rbp - 0x14], 0      | 0x123d | c6 45 ec 20          | mov byte ptr [rbp - |
| 3 0x1241 | 48 8b 15 c8 2d 00 00 | mov rdx, qword ptr [rip + 0x2dc8] | 0x1241 | 48 8b 15 c8 2d 00 00 | mov rdx, qword ptr  |

### Program 3 password\_check.o\_0x123d\_27 diassemblebly vs vanilla

|          |                      |                                   |        |                      |                     |
|----------|----------------------|-----------------------------------|--------|----------------------|---------------------|
| 0 0x1233 | b8 00 00 00 00       | mov eax, 0                        | 0x1233 | b8 00 00 00 00       | mov eax, 0          |
| 1 0x1238 | e8 a3 fe ff ff       | call 0x10e0                       | 0x1238 | e8 a3 fe ff ff       | call 0x10e0         |
| 2 0x123d | c6 45 ec 00          | mov byte ptr [rbp - 0x14], 0      | 0x123d | c6 45 ec 10          | mov byte ptr [rbp - |
| 3 0x1241 | 48 8b 15 c8 2d 00 00 | mov rdx, qword ptr [rip + 0x2dc8] | 0x1241 | 48 8b 15 c8 2d 00 00 | mov rdx, qword ptr  |

## Program 4 password\_check.o\_0x123d\_28 diassembleby vs vanilla

|          |                      |                                   |        |                      |                     |
|----------|----------------------|-----------------------------------|--------|----------------------|---------------------|
| 0 0x1233 | b8 00 00 00 00       | mov eax, 0                        | 0x1233 | b8 00 00 00 00       | mov eax, 0          |
| 1 0x1238 | e8 a3 fe ff ff       | call 0x10e0                       | 0x1238 | e8 a3 fe ff ff       | call 0x10e0         |
| 2 0x123d | c6 45 ec 00          | mov byte ptr [rbp - 0x14], 0      | 0x123d | c6 45 ec 08          | mov byte ptr [rbp - |
| 3 0x1241 | 48 8b 15 c8 2d 00 00 | mov rdx, qword ptr [rip + 0x2dc8] | 0x1241 | 48 8b 15 c8 2d 00 00 | mov rdx, qword ptr  |

### Program 5 password\_check.o\_0x123d\_29 diassembley vs vanilla

|          |                      |                                   |                             |                                   |
|----------|----------------------|-----------------------------------|-----------------------------|-----------------------------------|
| 0 0x1233 | b8 00 00 00 00       | mov eax, 0                        | 0x1233 b8 00 00 00 00       | mov eax, 0                        |
| 1 0x1238 | e8 a3 fe ff ff       | call 0x10e0                       | 0x1238 e8 a3 fe ff ff       | call 0x10e0                       |
| 2 0x123d | c6 45 ec 00          | mov byte ptr [rbp - 0x14], 0      | 0x123d c6 45 ec 04          | mov byte ptr [rbp - 0x14], 0      |
| 3 0x1241 | 48 8b 15 c8 2d 00 00 | mov rdx, qword ptr [rip + 0x2dc8] | 0x1241 48 8b 15 c8 2d 00 00 | mov rdx, qword ptr [rip + 0x2dc8] |

### Program 6 password\_check.o\_0x123d\_30 diassembley vs vanilla

|                               |                                   |                             |                     |
|-------------------------------|-----------------------------------|-----------------------------|---------------------|
| 0 0x1233 b8 00 00 00 00       | mov eax, 0                        | 0x1233 b8 00 00 00 00       | mov eax, 0          |
| 1 0x1238 e8 a3 fe ff ff       | call 0x10e0                       | 0x1238 e8 a3 fe ff ff       | call 0x10e0         |
| 2 0x123d c6 45 ec 00          | mov byte ptr [rbp - 0x14], 0      | 0x123d c6 45 ec 02          | mov byte ptr [rbp - |
| 3 0x1241 48 8b 15 c8 2d 00 00 | mov rdx, qword ptr [rip + 0x2dc8] | 0x1241 48 8b 15 c8 2d 00 00 | mov rdx, qword ptr  |

### Program 7 password\_check.o 0x123d 31 diassembley vs vanilla

|                               |                                   |                             |                     |
|-------------------------------|-----------------------------------|-----------------------------|---------------------|
| 0 0x1233 b8 00 00 00 00       | mov eax, 0                        | 0x1233 b8 00 00 00 00       | mov eax, 0          |
| 1 0x1238 e8 a3 fe ff ff       | call 0x10e0                       | 0x1238 e8 a3 fe ff ff       | call 0x10e0         |
| 2 0x123d c6 45 ec 00          | mov byte ptr [rbp - 0x14], 0      | 0x123d c6 45 ec 01          | mov byte ptr [rbp - |
| 3 0x1241 48 8b 15 c8 2d 00 00 | mov rdx, qword ptr [rip + 0x2dc8] | 0x1241 48 8b 15 c8 2d 00 00 | mov rdx, qword ptr  |

#### Program 8 password\_check.o\_0x1248\_29 diassemblebly vs vanilla

|                               |                                   |                             |                       |
|-------------------------------|-----------------------------------|-----------------------------|-----------------------|
| 0 0x1241 48 8b 15 c8 2d 00 00 | mov rdx, qword ptr [rip + 0x2dc8] | 0x1241 48 8b 15 c8 2d 00 00 | mov rdx, qword ptr    |
| 1 0x1248 48 8d 45 ed          | lea rax, [rbp - 0x13]             | 0x1248 48 8d 45 e9          | lea rax, [rbp - 0x13] |
| 2 0x124c be 0b 00 00 00       | mov esi, 0xb                      | 0x124c be 0b 00 00 00       | mov esi, 0xb          |
| 3 0x1251 48 89 c7             | mov rdi, rax                      | 0x1251 48 89 c7             | mov rdi, rax          |

#### Program 9 password\_check.o\_0x1248\_31 diassemblebly vs vanilla

|                               |                                   |                             |                       |
|-------------------------------|-----------------------------------|-----------------------------|-----------------------|
| 0 0x1241 48 8b 15 c8 2d 00 00 | mov rdx, qword ptr [rip + 0x2dc8] | 0x1241 48 8b 15 c8 2d 00 00 | mov rdx, qword ptr    |
| 1 0x1248 48 8d 45 ed          | lea rax, [rbp - 0x13]             | 0x1248 48 8d 45 ec          | lea rax, [rbp - 0x13] |
| 2 0x124c be 0b 00 00 00       | mov esi, 0xb                      | 0x124c be 0b 00 00 00       | mov esi, 0xb          |
| 3 0x1251 48 89 c7             | mov rdi, rax                      | 0x1251 48 89 c7             | mov rdi, rax          |

#### Program 10 password\_check.o\_0x12a0\_19 diassemblebly vs vanilla

|                               |                        |                             |                        |  |
|-------------------------------|------------------------|-----------------------------|------------------------|--|
| 0 0x1299 48 8d 15 93 0d 00 00 | lea rdx, [rip + 0xd93] | 0x1299 48 8d 15 93 0d 00 00 | lea rdx, [rip + 0xd93] |  |
| 1 0x12a0 48 89 d6             | mov rsi, rdx           | 0x12a0 48 89 c6             | mov rsi, rax           |  |
| 2 0x12a3 48 89 c7             | mov rdi, rax           | 0x12a3 48 89 c7             | mov rdi, rax           |  |
| 3 0x12a6 e8 65 fe ff ff       | call 0x1110            | 0x12a6 e8 65 fe ff ff       | call 0x1110            |  |

#### Program 11 password\_check.o\_0x12a3\_19 diassemblebly vs vanilla

|                               |                        |                             |                        |  |
|-------------------------------|------------------------|-----------------------------|------------------------|--|
| 0 0x1299 48 8d 15 93 0d 00 00 | lea rdx, [rip + 0xd93] | 0x1299 48 8d 15 93 0d 00 00 | lea rdx, [rip + 0xd93] |  |
| 1 0x12a0 48 89 d6             | mov rsi, rdx           | 0x12a0 48 89 d6             | mov rsi, rdx           |  |
| 2 0x12a3 48 89 c7             | mov rdi, rax           | 0x12a3 48 89 d7             | mov rdi, rdx           |  |
| 3 0x12a6 e8 65 fe ff ff       | call 0x1110            | 0x12a6 e8 65 fe ff ff       | call 0x1110            |  |
| 4 0x12ab 85 c0                | test eax, eax          | 0x12ab 85 c0                | test eax, eax          |  |
| 5 0x12ad 75 1a                | jne 0x12c9             | 0x12ad 75 1a                | jne 0x12c9             |  |

#### Program 12 password\_check.o\_0x12a3\_23 diassemblebly vs vanilla

|                               |                        |                             |                        |  |
|-------------------------------|------------------------|-----------------------------|------------------------|--|
| 0 0x1299 48 8d 15 93 0d 00 00 | lea rdx, [rip + 0xd93] | 0x1299 48 8d 15 93 0d 00 00 | lea rdx, [rip + 0xd93] |  |
| 1 0x12a0 48 89 d6             | mov rsi, rdx           | 0x12a0 48 89 d6             | mov rsi, rdx           |  |
| 2 0x12a3 48 89 c7             | mov rdi, rax           | 0x12a3 48 89 c6             | mov rsi, rax           |  |
| 3 0x12a6 e8 65 fe ff ff       | call 0x1110            | 0x12a6 e8 65 fe ff ff       | call 0x1110            |  |
| 4 0x12ab 85 c0                | test eax, eax          | 0x12ab 85 c0                | test eax, eax          |  |
| 5 0x12ad 75 1a                | jne 0x12c9             | 0x12ad 75 1a                | jne 0x12c9             |  |

#### Program 13 password\_check.o\_0x12ab\_10 diassemblebly vs vanilla

|                               |                              |                             |                              |
|-------------------------------|------------------------------|-----------------------------|------------------------------|
| 0 0x12a3 48 89 c7             | mov rdi, rax                 | 0x12a3 48 89 c7             | mov rdi, rax                 |
| 1 0x12a6 e8 65 fe ff ff       | call 0x1110                  | 0x12a6 e8 65 fe ff ff       | call 0x1110                  |
| 2 0x12ab 85 c0                | test eax, eax                | 0x12ab 85 e0                | test eax, esp                |
| 3 0x12ad 75 1a                | jne 0x12c9                   | 0x12ad 75 1a                | jne 0x12c9                   |
| 4 0x12af c6 45 ec 01          | mov byte ptr [rbp - 0x14], 1 | 0x12af c6 45 ec 01          | mov byte ptr [rbp - 0x14], 1 |
| 5 0x12b3 eb 14                | jmp 0x12c9                   | 0x12b3 eb 14                | jmp 0x12c9                   |
| 6 0x12b5 48 8d 05 7c 0d 00 00 | lea rax, [rip + 0xd7c]       | 0x12b5 48 8d 05 7c 0d 00 00 | lea rax, [rip + 0xd7c]       |

#### Program 14 password\_check.o\_0x12ab\_11 diassemblebly vs vanilla

|                               |                              |                             |                              |
|-------------------------------|------------------------------|-----------------------------|------------------------------|
| 0 0x12a3 48 89 c7             | mov rdi, rax                 | 0x12a3 48 89 c7             | mov rdi, rax                 |
| 1 0x12a6 e8 65 fe ff ff       | call 0x1110                  | 0x12a6 e8 65 fe ff ff       | call 0x1110                  |
| 2 0x12ab 85 c0                | test eax, eax                | 0x12ab 85 d0                | test eax, edx                |
| 3 0x12ad 75 1a                | jne 0x12c9                   | 0x12ad 75 1a                | jne 0x12c9                   |
| 4 0x12af c6 45 ec 01          | mov byte ptr [rbp - 0x14], 1 | 0x12af c6 45 ec 01          | mov byte ptr [rbp - 0x14], 1 |
| 5 0x12b3 eb 14                | jmp 0x12c9                   | 0x12b3 eb 14                | jmp 0x12c9                   |
| 6 0x12b5 48 8d 05 7c 0d 00 00 | lea rax, [rip + 0xd7c]       | 0x12b5 48 8d 05 7c 0d 00 00 | lea rax, [rip + 0xd7c]       |

#### Program 15 password\_check.o\_0x12ab\_13 diassemblebly vs vanilla

|                               |                              |                             |                              |
|-------------------------------|------------------------------|-----------------------------|------------------------------|
| 0 0x12a3 48 89 c7             | mov rdi, rax                 | 0x12a3 48 89 c7             | mov rdi, rax                 |
| 1 0x12a6 e8 65 fe ff ff       | call 0x1110                  | 0x12a6 e8 65 fe ff ff       | call 0x1110                  |
| 2 0x12ab 85 c0                | test eax, eax                | 0x12ab 85 c4                | test esp, eax                |
| 3 0x12ad 75 1a                | jne 0x12c9                   | 0x12ad 75 1a                | jne 0x12c9                   |
| 4 0x12af c6 45 ec 01          | mov byte ptr [rbp - 0x14], 1 | 0x12af c6 45 ec 01          | mov byte ptr [rbp - 0x14], 1 |
| 5 0x12b3 eb 14                | jmp 0x12c9                   | 0x12b3 eb 14                | jmp 0x12c9                   |
| 6 0x12b5 48 8d 05 7c 0d 00 00 | lea rax, [rip + 0xd7c]       | 0x12b5 48 8d 05 7c 0d 00 00 | lea rax, [rip + 0xd7c]       |

#### Program 16 password\_check.o\_0x12ab\_14 diassemblebly vs vanilla

|                               |                              |                             |                              |
|-------------------------------|------------------------------|-----------------------------|------------------------------|
| 0 0x12a3 48 89 c7             | mov rdi, rax                 | 0x12a3 48 89 c7             | mov rdi, rax                 |
| 1 0x12a6 e8 65 fe ff ff       | call 0x1110                  | 0x12a6 e8 65 fe ff ff       | call 0x1110                  |
| 2 0x12ab 85 c0                | test eax, eax                | 0x12ab 85 c2                | test edx, eax                |
| 3 0x12ad 75 1a                | jne 0x12c9                   | 0x12ad 75 1a                | jne 0x12c9                   |
| 4 0x12af c6 45 ec 01          | mov byte ptr [rbp - 0x14], 1 | 0x12af c6 45 ec 01          | mov byte ptr [rbp - 0x14], 1 |
| 5 0x12b3 eb 14                | jmp 0x12c9                   | 0x12b3 eb 14                | jmp 0x12c9                   |
| 6 0x12b5 48 8d 05 7c 0d 00 00 | lea rax, [rip + 0xd7c]       | 0x12b5 48 8d 05 7c 0d 00 00 | lea rax, [rip + 0xd7c]       |

#### Program 17 password\_check.o\_0x12ad\_7 diassemblebly vs vanilla

|                               |                              |                             |                              |
|-------------------------------|------------------------------|-----------------------------|------------------------------|
| 0 0x12a3 48 89 c7             | mov rdi, rax                 | 0x12a3 48 89 c7             | mov rdi, rax                 |
| 1 0x12a6 e8 65 fe ff ff       | call 0x1110                  | 0x12a6 e8 65 fe ff ff       | call 0x1110                  |
| 2 0x12ab 85 c0                | test eax, eax                | 0x12ab 85 c0                | test eax, eax                |
| 3 0x12ad 75 1a                | jne 0x12c9                   | 0x12ad 74 1a                | je 0x12c9                    |
| 4 0x12af c6 45 ec 01          | mov byte ptr [rbp - 0x14], 1 | 0x12af c6 45 ec 01          | mov byte ptr [rbp - 0x14], 1 |
| 5 0x12b3 eb 14                | jmp 0x12c9                   | 0x12b3 eb 14                | jmp 0x12c9                   |
| 6 0x12b5 48 8d 05 7c 0d 00 00 | lea rax, [rip + 0xd7c]       | 0x12b5 48 8d 05 7c 0d 00 00 | lea rax, [rip + 0xd7c]       |

#### Program 18 password\_check.o\_0x12ad\_13 diassemblebly vs vanilla

|                               |                              |                             |                              |
|-------------------------------|------------------------------|-----------------------------|------------------------------|
| 0 0x12a3 48 89 c7             | mov rdi, rax                 | 0x12a3 48 89 c7             | mov rdi, rax                 |
| 1 0x12a6 e8 65 fe ff ff       | call 0x1110                  | 0x12a6 e8 65 fe ff ff       | call 0x1110                  |
| 2 0x12ab 85 c0                | test eax, eax                | 0x12ab 85 c0                | test eax, eax                |
| 3 0x12ad 75 1a                | jne 0x12c9                   | 0x12ad 75 1e                | jne 0x12cd                   |
| 4 0x12af c6 45 ec 01          | mov byte ptr [rbp - 0x14], 1 | 0x12af c6 45 ec 01          | mov byte ptr [rbp - 0x14], 1 |
| 5 0x12b3 eb 14                | jmp 0x12c9                   | 0x12b3 eb 14                | jmp 0x12c9                   |
| 6 0x12b5 48 8d 05 7c 0d 00 00 | lea rax, [rip + 0xd7c]       | 0x12b5 48 8d 05 7c 0d 00 00 | lea rax, [rip + 0xd7c]       |

#### Program 19 password\_check.o\_0x12a6\_9 diassemblebly vs vanilla

|                         |                              |                       |                              |  |
|-------------------------|------------------------------|-----------------------|------------------------------|--|
| 0 0x12a0 48 89 d6       | mov rsi, rdx                 | 0x12a0 48 89 d6       | mov rsi, rdx                 |  |
| 1 0x12a3 48 89 c7       | mov rdi, rax                 | 0x12a3 48 89 c7       | mov rdi, rax                 |  |
| 2 0x12a6 e8 65 fe ff ff | call 0x1110                  | 0x12a6 e8 25 fe ff ff | call 0x10d0                  |  |
| 3 0x12ab 85 c0          | test eax, eax                | 0x12ab 85 c0          | test eax, eax                |  |
| 4 0x12ad 75 1a          | jne 0x12c9                   | 0x12ad 75 1a          | jne 0x12c9                   |  |
| 5 0x12af c6 45 ec 01    | mov byte ptr [rbp - 0x14], 1 | 0x12af c6 45 ec 01    | mov byte ptr [rbp - 0x14], 1 |  |

#### Program 20 password\_check.o\_0x12c9\_1 diassemblebly vs vanilla

|                               |                              |                             |                              |
|-------------------------------|------------------------------|-----------------------------|------------------------------|
| 0 0x12bf b8 00 00 00 00       | mov eax, 0                   | 0x12bf b8 00 00 00 00       | mov eax, 0                   |
| 1 0x12c4 e8 17 fe ff ff       | call 0x10e0                  | 0x12c4 e8 17 fe ff ff       | call 0x10e0                  |
| 2 0x12c9 80 7d ec 00          | cmp byte ptr [rbp - 0x14], 0 | 0x12c9 c0 7d ec 00          | sar byte ptr [rbp - 0x14], 1 |
| 3 0x12cd 74 16                | je 0x12e5                    | 0x12cd 74 16                | je 0x12e5                    |
| 4 0x12cf 48 8d 05 6b 0d 00 00 | lea rax, [rip + 0xd6b]       | 0x12cf 48 8d 05 6b 0d 00 00 | lea rax, [rip + 0xd6b]       |

#### Program 21 password\_check.o\_0x12c9\_14 diassemblebly vs vanilla

|                         |                              |                       |                              |
|-------------------------|------------------------------|-----------------------|------------------------------|
| 0 0x12bf b8 00 00 00 00 | mov eax, 0                   | 0x12bf b8 00 00 00 00 | mov eax, 0                   |
| 1 0x12c4 e8 17 fe ff ff | call 0x10e0                  | 0x12c4 e8 17 fe ff ff | call 0x10e0                  |
| 2 0x12c9 80 7d ec 00    | cmp byte ptr [rbp - 0x14], 0 | 0x12c9 80 7f ec 00    | cmp byte ptr [rdi - 0x14], 0 |
| 3 0x12cd 74 16          | je 0x12e5                    | 0x12cd 74 16          | je 0x12e5                    |

|                               |                        |                             |                        |
|-------------------------------|------------------------|-----------------------------|------------------------|
| 4 0x12cf 48 8d 05 6b 0d 00 00 | lea rax, [rip + 0xd6b] | 0x12cf 48 8d 05 6b 0d 00 00 | lea rax, [rip + 0xd6b] |
|-------------------------------|------------------------|-----------------------------|------------------------|

**Program 22 password\_check.o\_0x12c9\_16 diassemblebly vs vanilla**

|                               |                              |                             |                           |
|-------------------------------|------------------------------|-----------------------------|---------------------------|
| 0 0x12bf b8 00 00 00 00       | mov eax, 0                   | 0x12bf b8 00 00 00 00       | mov eax, 0                |
| 1 0x12c4 e8 17 fe ff ff       | call 0x10e0                  | 0x12c4 e8 17 fe ff ff       | call 0x10e0               |
| 2 0x12c9 80 7d ec 00          | cmp byte ptr [rbp - 0x14], 0 | 0x12c9 80 7d 6c 00          | cmp byte ptr [rbp + 0x6c] |
| 3 0x12cd 74 16                | je 0x12e5                    | 0x12cd 74 16                | je 0x12e5                 |
| 4 0x12cf 48 8d 05 6b 0d 00 00 | lea rax, [rip + 0xd6b]       | 0x12cf 48 8d 05 6b 0d 00 00 | lea rax, [rip + 0xd6b]    |

**Program 23 password\_check.o\_0x12c9\_17 diassemblebly vs vanilla**

|                               |                              |                             |                           |
|-------------------------------|------------------------------|-----------------------------|---------------------------|
| 0 0x12bf b8 00 00 00 00       | mov eax, 0                   | 0x12bf b8 00 00 00 00       | mov eax, 0                |
| 1 0x12c4 e8 17 fe ff ff       | call 0x10e0                  | 0x12c4 e8 17 fe ff ff       | call 0x10e0               |
| 2 0x12c9 80 7d ec 00          | cmp byte ptr [rbp - 0x14], 0 | 0x12c9 80 7d ac 00          | cmp byte ptr [rbp - 0x54] |
| 3 0x12cd 74 16                | je 0x12e5                    | 0x12cd 74 16                | je 0x12e5                 |
| 4 0x12cf 48 8d 05 6b 0d 00 00 | lea rax, [rip + 0xd6b]       | 0x12cf 48 8d 05 6b 0d 00 00 | lea rax, [rip + 0xd6b]    |

**Program 24 password\_check.o\_0x12c9\_18 diassemblebly vs vanilla**

|                               |                              |                             |                           |
|-------------------------------|------------------------------|-----------------------------|---------------------------|
| 0 0x12bf b8 00 00 00 00       | mov eax, 0                   | 0x12bf b8 00 00 00 00       | mov eax, 0                |
| 1 0x12c4 e8 17 fe ff ff       | call 0x10e0                  | 0x12c4 e8 17 fe ff ff       | call 0x10e0               |
| 2 0x12c9 80 7d ec 00          | cmp byte ptr [rbp - 0x14], 0 | 0x12c9 80 7d cc 00          | cmp byte ptr [rbp - 0x34] |
| 3 0x12cd 74 16                | je 0x12e5                    | 0x12cd 74 16                | je 0x12e5                 |
| 4 0x12cf 48 8d 05 6b 0d 00 00 | lea rax, [rip + 0xd6b]       | 0x12cf 48 8d 05 6b 0d 00 00 | lea rax, [rip + 0xd6b]    |

**Program 25 password\_check.o\_0x12c9\_19 diassemblebly vs vanilla**

|                               |                              |                             |                           |
|-------------------------------|------------------------------|-----------------------------|---------------------------|
| 0 0x12bf b8 00 00 00 00       | mov eax, 0                   | 0x12bf b8 00 00 00 00       | mov eax, 0                |
| 1 0x12c4 e8 17 fe ff ff       | call 0x10e0                  | 0x12c4 e8 17 fe ff ff       | call 0x10e0               |
| 2 0x12c9 80 7d ec 00          | cmp byte ptr [rbp - 0x14], 0 | 0x12c9 80 7d fc 00          | cmp byte ptr [rbp - 4], 0 |
| 3 0x12cd 74 16                | je 0x12e5                    | 0x12cd 74 16                | je 0x12e5                 |
| 4 0x12cf 48 8d 05 6b 0d 00 00 | lea rax, [rip + 0xd6b]       | 0x12cf 48 8d 05 6b 0d 00 00 | lea rax, [rip + 0xd6b]    |

**Program 26 password\_check.o\_0x12c9\_20 diassemblebly vs vanilla**

|                               |                              |                             |                           |
|-------------------------------|------------------------------|-----------------------------|---------------------------|
| 0 0x12bf b8 00 00 00 00       | mov eax, 0                   | 0x12bf b8 00 00 00 00       | mov eax, 0                |
| 1 0x12c4 e8 17 fe ff ff       | call 0x10e0                  | 0x12c4 e8 17 fe ff ff       | call 0x10e0               |
| 2 0x12c9 80 7d ec 00          | cmp byte ptr [rbp - 0x14], 0 | 0x12c9 80 7d e4 00          | cmp byte ptr [rbp - 0x1c] |
| 3 0x12cd 74 16                | je 0x12e5                    | 0x12cd 74 16                | je 0x12e5                 |
| 4 0x12cf 48 8d 05 6b 0d 00 00 | lea rax, [rip + 0xd6b]       | 0x12cf 48 8d 05 6b 0d 00 00 | lea rax, [rip + 0xd6b]    |

**Program 27 password\_check.o\_0x12c9\_21 diassemblebly vs vanilla**

|                               |                              |                             |                           |
|-------------------------------|------------------------------|-----------------------------|---------------------------|
| 0 0x12bf b8 00 00 00 00       | mov eax, 0                   | 0x12bf b8 00 00 00 00       | mov eax, 0                |
| 1 0x12c4 e8 17 fe ff ff       | call 0x10e0                  | 0x12c4 e8 17 fe ff ff       | call 0x10e0               |
| 2 0x12c9 80 7d ec 00          | cmp byte ptr [rbp - 0x14], 0 | 0x12c9 80 7d e8 00          | cmp byte ptr [rbp - 0x18] |
| 3 0x12cd 74 16                | je 0x12e5                    | 0x12cd 74 16                | je 0x12e5                 |
| 4 0x12cf 48 8d 05 6b 0d 00 00 | lea rax, [rip + 0xd6b]       | 0x12cf 48 8d 05 6b 0d 00 00 | lea rax, [rip + 0xd6b]    |

**Program 28 password\_check.o\_0x12c9\_22 diassemblebly vs vanilla**

|                               |                              |                             |                           |
|-------------------------------|------------------------------|-----------------------------|---------------------------|
| 0 0x12bf b8 00 00 00 00       | mov eax, 0                   | 0x12bf b8 00 00 00 00       | mov eax, 0                |
| 1 0x12c4 e8 17 fe ff ff       | call 0x10e0                  | 0x12c4 e8 17 fe ff ff       | call 0x10e0               |
| 2 0x12c9 80 7d ec 00          | cmp byte ptr [rbp - 0x14], 0 | 0x12c9 80 7d ee 00          | cmp byte ptr [rbp - 0x12] |
| 3 0x12cd 74 16                | je 0x12e5                    | 0x12cd 74 16                | je 0x12e5                 |
| 4 0x12cf 48 8d 05 6b 0d 00 00 | lea rax, [rip + 0xd6b]       | 0x12cf 48 8d 05 6b 0d 00 00 | lea rax, [rip + 0xd6b]    |

**Program 29 password\_check.o\_0x12c9\_23 diassemblebly vs vanilla**

|                         |                              |                       |                           |
|-------------------------|------------------------------|-----------------------|---------------------------|
| 0 0x12bf b8 00 00 00 00 | mov eax, 0                   | 0x12bf b8 00 00 00 00 | mov eax, 0                |
| 1 0x12c4 e8 17 fe ff ff | call 0x10e0                  | 0x12c4 e8 17 fe ff ff | call 0x10e0               |
| 2 0x12c9 80 7d ec 00    | cmp byte ptr [rbp - 0x14], 0 | 0x12c9 80 7d ed 00    | cmp byte ptr [rbp - 0x13] |

|                               |                        |                             |                        |
|-------------------------------|------------------------|-----------------------------|------------------------|
| 3 0x12cd 74 16                | je 0x12e5              | 0x12cd 74 16                | je 0x12e5              |
| 4 0x12cf 48 8d 05 6b 0d 00 00 | lea rax, [rip + 0xd6b] | 0x12cf 48 8d 05 6b 0d 00 00 | lea rax, [rip + 0xd6b] |

### Program 30 password\_check.o\_0x12c9\_24 diassembley vs vanilla

|                               |                              |                             |                           |
|-------------------------------|------------------------------|-----------------------------|---------------------------|
| 0 0x12bf b8 00 00 00 00       | mov eax, 0                   | 0x12bf b8 00 00 00 00       | mov eax, 0                |
| 1 0x12c4 e8 17 fe ff ff       | call 0x10e0                  | 0x12c4 e8 17 fe ff ff       | call 0x10e0               |
| 2 0x12c9 80 7d ec 00          | cmp byte ptr [rbp - 0x14], 0 | 0x12c9 80 7d ec 80          | cmp byte ptr [rbp - 0x14] |
| 3 0x12cd 74 16                | je 0x12e5                    | 0x12cd 74 16                | je 0x12e5                 |
| 4 0x12cf 48 8d 05 6b 0d 00 00 | lea rax, [rip + 0xd6b]       | 0x12cf 48 8d 05 6b 0d 00 00 | lea rax, [rip + 0xd6b]    |

### Program 31 password\_check.o\_0x12c9\_25 diassembley vs vanilla

|                               |                              |                             |                           |
|-------------------------------|------------------------------|-----------------------------|---------------------------|
| 0 0x12bf b8 00 00 00 00       | mov eax, 0                   | 0x12bf b8 00 00 00 00       | mov eax, 0                |
| 1 0x12c4 e8 17 fe ff ff       | call 0x10e0                  | 0x12c4 e8 17 fe ff ff       | call 0x10e0               |
| 2 0x12c9 80 7d ec 00          | cmp byte ptr [rbp - 0x14], 0 | 0x12c9 80 7d ec 40          | cmp byte ptr [rbp - 0x14] |
| 3 0x12cd 74 16                | je 0x12e5                    | 0x12cd 74 16                | je 0x12e5                 |
| 4 0x12cf 48 8d 05 6b 0d 00 00 | lea rax, [rip + 0xd6b]       | 0x12cf 48 8d 05 6b 0d 00 00 | lea rax, [rip + 0xd6b]    |

### Program 32 password\_check.o\_0x12c9\_26 diassembley vs vanilla

|                               |                              |                             |                           |
|-------------------------------|------------------------------|-----------------------------|---------------------------|
| 0 0x12bf b8 00 00 00 00       | mov eax, 0                   | 0x12bf b8 00 00 00 00       | mov eax, 0                |
| 1 0x12c4 e8 17 fe ff ff       | call 0x10e0                  | 0x12c4 e8 17 fe ff ff       | call 0x10e0               |
| 2 0x12c9 80 7d ec 00          | cmp byte ptr [rbp - 0x14], 0 | 0x12c9 80 7d ec 20          | cmp byte ptr [rbp - 0x14] |
| 3 0x12cd 74 16                | je 0x12e5                    | 0x12cd 74 16                | je 0x12e5                 |
| 4 0x12cf 48 8d 05 6b 0d 00 00 | lea rax, [rip + 0xd6b]       | 0x12cf 48 8d 05 6b 0d 00 00 | lea rax, [rip + 0xd6b]    |

### Program 33 password\_check.o 0x12c9\_27 diassembley vs vanilla

|                               |                              |                             |                           |
|-------------------------------|------------------------------|-----------------------------|---------------------------|
| 0 0x12bf b8 00 00 00 00       | mov eax, 0                   | 0x12bf b8 00 00 00 00       | mov eax, 0                |
| 1 0x12c4 e8 17 fe ff ff       | call 0x10e0                  | 0x12c4 e8 17 fe ff ff       | call 0x10e0               |
| 2 0x12c9 80 7d ec 00          | cmp byte ptr [rbp - 0x14], 0 | 0x12c9 80 7d ec 10          | cmp byte ptr [rbp - 0x14] |
| 3 0x12cd 74 16                | je 0x12e5                    | 0x12cd 74 16                | je 0x12e5                 |
| 4 0x12cf 48 8d 05 6b 0d 00 00 | lea rax, [rip + 0xd6b]       | 0x12cf 48 8d 05 6b 0d 00 00 | lea rax, [rip + 0xd6b]    |

### Program 34 password\_check.o\_0x12c9\_28 diassembley vs vanilla

|                               |                              |                             |                           |
|-------------------------------|------------------------------|-----------------------------|---------------------------|
| 0 0x12bf b8 00 00 00 00       | mov eax, 0                   | 0x12bf b8 00 00 00 00       | mov eax, 0                |
| 1 0x12c4 e8 17 fe ff ff       | call 0x10e0                  | 0x12c4 e8 17 fe ff ff       | call 0x10e0               |
| 2 0x12c9 80 7d ec 00          | cmp byte ptr [rbp - 0x14], 0 | 0x12c9 80 7d ec 08          | cmp byte ptr [rbp - 0x14] |
| 3 0x12cd 74 16                | je 0x12e5                    | 0x12cd 74 16                | je 0x12e5                 |
| 4 0x12cf 48 8d 05 6b 0d 00 00 | lea rax, [rip + 0xd6b]       | 0x12cf 48 8d 05 6b 0d 00 00 | lea rax, [rip + 0xd6b]    |

### Program 35 password\_check.o\_0x12c9\_29 diassembley vs vanilla

|                               |                              |                             |                           |
|-------------------------------|------------------------------|-----------------------------|---------------------------|
| 0 0x12bf b8 00 00 00 00       | mov eax, 0                   | 0x12bf b8 00 00 00 00       | mov eax, 0                |
| 1 0x12c4 e8 17 fe ff ff       | call 0x10e0                  | 0x12c4 e8 17 fe ff ff       | call 0x10e0               |
| 2 0x12c9 80 7d ec 00          | cmp byte ptr [rbp - 0x14], 0 | 0x12c9 80 7d ec 04          | cmp byte ptr [rbp - 0x14] |
| 3 0x12cd 74 16                | je 0x12e5                    | 0x12cd 74 16                | je 0x12e5                 |
| 4 0x12cf 48 8d 05 6b 0d 00 00 | lea rax, [rip + 0xd6b]       | 0x12cf 48 8d 05 6b 0d 00 00 | lea rax, [rip + 0xd6b]    |

### Program 36 password\_check.o\_0x12c9\_30 diassembley vs vanilla

|                               |                              |                             |                           |
|-------------------------------|------------------------------|-----------------------------|---------------------------|
| 0 0x12bf b8 00 00 00 00       | mov eax, 0                   | 0x12bf b8 00 00 00 00       | mov eax, 0                |
| 1 0x12c4 e8 17 fe ff ff       | call 0x10e0                  | 0x12c4 e8 17 fe ff ff       | call 0x10e0               |
| 2 0x12c9 80 7d ec 00          | cmp byte ptr [rbp - 0x14], 0 | 0x12c9 80 7d ec 02          | cmp byte ptr [rbp - 0x14] |
| 3 0x12cd 74 16                | je 0x12e5                    | 0x12cd 74 16                | je 0x12e5                 |
| 4 0x12cf 48 8d 05 6b 0d 00 00 | lea rax, [rip + 0xd6b]       | 0x12cf 48 8d 05 6b 0d 00 00 | lea rax, [rip + 0xd6b]    |

### Program 37 password\_check.o\_0x12c9\_31 diassemblebly vs vanilla

|                               |                              |                             |                           |
|-------------------------------|------------------------------|-----------------------------|---------------------------|
| 0 0x12bf b8 00 00 00 00       | mov eax, 0                   | 0x12bf b8 00 00 00 00       | mov eax, 0                |
| 1 0x12c4 e8 17 fe ff ff       | call 0x10e0                  | 0x12c4 e8 17 fe ff ff       | call 0x10e0               |
| 2 0x12c9 80 7d ec 00          | cmp byte ptr [rbp - 0x14], 0 | 0x12c9 80 7d ec 01          | cmp byte ptr [rbp - 0x14] |
| 3 0x12cd 74 16                | je 0x12e5                    | 0x12cd 74 16                | je 0x12e5                 |
| 4 0x12cf 48 8d 05 6b 0d 00 00 | lea rax, [rip + 0xd6b]       | 0x12cf 48 8d 05 6b 0d 00 00 | lea rax, [rip + 0xd6b]    |

### Program 38 password\_check.o\_0x12cd\_1 diassemblebly vs vanilla

|                               |                              |                             |                           |
|-------------------------------|------------------------------|-----------------------------|---------------------------|
| 0 0x12c4 e8 17 fe ff ff       | call 0x10e0                  | 0x12c4 e8 17 fe ff ff       | call 0x10e0               |
| 1 0x12c9 80 7d ec 00          | cmp byte ptr [rbp - 0x14], 0 | 0x12c9 80 7d ec 00          | cmp byte ptr [rbp - 0x14] |
| 2 0x12cd 74 16                | je 0x12e5                    | 0x12cd 34 16                | xor al, 0x16              |
| 3 0x12cf 48 8d 05 6b 0d 00 00 | lea rax, [rip + 0xd6b]       | 0x12cf 48 8d 05 6b 0d 00 00 | lea rax, [rip + 0xd6b]    |
| 4 0x12d6 48 89 c7             | mov rdi, rax                 | 0x12d6 48 89 c7             | mov rdi, rax              |

### Program 39 password\_check.o\_0x12cd\_4 diassemblebly vs vanilla

|                               |                              |                             |                           |
|-------------------------------|------------------------------|-----------------------------|---------------------------|
| 0 0x12c4 e8 17 fe ff ff       | call 0x10e0                  | 0x12c4 e8 17 fe ff ff       | call 0x10e0               |
| 1 0x12c9 80 7d ec 00          | cmp byte ptr [rbp - 0x14], 0 | 0x12c9 80 7d ec 00          | cmp byte ptr [rbp - 0x14] |
| 2 0x12cd 74 16                | je 0x12e5                    | 0x12cd 7c 16                | j1 0x12e5                 |
| 3 0x12cf 48 8d 05 6b 0d 00 00 | lea rax, [rip + 0xd6b]       | 0x12cf 48 8d 05 6b 0d 00 00 | lea rax, [rip + 0xd6b]    |
| 4 0x12d6 48 89 c7             | mov rdi, rax                 | 0x12d6 48 89 c7             | mov rdi, rax              |

### Program 40 password\_check.o\_0x12cd\_5 diassemblebly vs vanilla

|                               |                              |                             |                           |
|-------------------------------|------------------------------|-----------------------------|---------------------------|
| 0 0x12c4 e8 17 fe ff ff       | call 0x10e0                  | 0x12c4 e8 17 fe ff ff       | call 0x10e0               |
| 1 0x12c9 80 7d ec 00          | cmp byte ptr [rbp - 0x14], 0 | 0x12c9 80 7d ec 00          | cmp byte ptr [rbp - 0x14] |
| 2 0x12cd 74 16                | je 0x12e5                    | 0x12cd 70 16                | jo 0x12e5                 |
| 3 0x12cf 48 8d 05 6b 0d 00 00 | lea rax, [rip + 0xd6b]       | 0x12cf 48 8d 05 6b 0d 00 00 | lea rax, [rip + 0xd6b]    |
| 4 0x12d6 48 89 c7             | mov rdi, rax                 | 0x12d6 48 89 c7             | mov rdi, rax              |

### Program 41 password\_check.o\_0x12cd\_7 diassemblebly vs vanilla

|                               |                              |                             |                           |
|-------------------------------|------------------------------|-----------------------------|---------------------------|
| 0 0x12c4 e8 17 fe ff ff       | call 0x10e0                  | 0x12c4 e8 17 fe ff ff       | call 0x10e0               |
| 1 0x12c9 80 7d ec 00          | cmp byte ptr [rbp - 0x14], 0 | 0x12c9 80 7d ec 00          | cmp byte ptr [rbp - 0x14] |
| 2 0x12cd 74 16                | je 0x12e5                    | 0x12cd 75 16                | jne 0x12e5                |
| 3 0x12cf 48 8d 05 6b 0d 00 00 | lea rax, [rip + 0xd6b]       | 0x12cf 48 8d 05 6b 0d 00 00 | lea rax, [rip + 0xd6b]    |
| 4 0x12d6 48 89 c7             | mov rdi, rax                 | 0x12d6 48 89 c7             | mov rdi, rax              |

### Program 42 password\_check.o\_0x12e5\_28 diassemblebly vs vanilla

|                               |                        |                             |                        |  |
|-------------------------------|------------------------|-----------------------------|------------------------|--|
| 0 0x12de b8 00 00 00 00       | mov eax, 0             | 0x12de b8 00 00 00 00       | mov eax, 0             |  |
| 1 0x12e3 eb 14                | jmp 0x12f9             | 0x12e3 eb 14                | jmp 0x12f9             |  |
| 2 0x12e5 48 8d 05 5d 0d 00 00 | lea rax, [rip + 0xd5d] | 0x12e5 48 8d 05 55 0d 00 00 | lea rax, [rip + 0xd55] |  |
| 3 0x12ec 48 89 c7             | mov rdi, rax           | 0x12ec 48 89 c7             | mov rdi, rax           |  |
| 4 0x12ef e8 bc fd ff ff       | call 0x10b0            | 0x12ef e8 bc fd ff ff       | call 0x10b0            |  |

### Source Code Lines

```
#include <stdio.h>
#include <stdlib.h> // Required for the exit() function
#include <stdbool.h>
#include <string.h>
// Below provides "EXIT_SUCCESS"
#include <stdbool.h>

// Correct password:
#define PASSWORD "pass"

// Password input max length:
#define MAX_LENGTH 10
```

```

/*
 * int main
 * A sample program to that has a user enter a password and sees if it's correct
 */
int main() {

    // Buffer to store user input (extra byte for null terminator)
    char input[MAX_LENGTH + 1];
    printf("Enter the password (max %d characters): ", MAX_LENGTH);

    bool password_correct = false;

    // fgets takes arguments: (buffer, buffer_size, input)
    // this assures that the input is no longer than the size of the buffer
    if (fgets(input, sizeof(input), stdin) != NULL) {
        // If the entered password exceeds the
        // buffer it's incorrect
        if (strchr(input, '\n') == NULL) {
            password_correct = false;
        }
        else {
            // Remove the newling character
            input[strcspn(input, "\n")] = '\0';

            // NOTICE: fgets and strchr make sure the password
            // is the correct length so strcmp is safe :)
            if (strcmp(input, PASSWORD) == 0) {
                password_correct = true;
            }
        }
    }
    else {
        printf("no input");
    }

    // Compare the input with the predefined password
    if (password_correct == 1) {
        printf("Correct\n");
        // Exit with 0 when exit is correct!
        return EXIT_SUCCESS;
    } else {
        printf("Wrong\n");
        return 97;
    }

    return 84;
}

```